Referencias científicas

[1] Artículo principal

Horowitz, S. L., & Pavlidis, T. (1976). Picture Segmentation by a Tree Traversal Algorithm. Journal of the ACM, 23(2), 368-388.

https://doi.org/10.1145/321941.321956

[2] Aplicación complementaria

Fu, G., Zhao, H., Li, C., & Shi, L. (2013). Segmentation for High-Resolution Optical Remote Sensing Imagery Using Improved Quadtree and Region Adjacency Graph Technique. Remote Sensing, 5(7), 3259-3279.

https://doi.org/10.3390/rs5073259

[3] Otsu

Otsu, N. (1979). A Threshold Selection Method from Gray-Level Histograms. IEEE Transactions on Systems, Man, and Cybernetics, 9(1), 62-66.

https://doi.org/10.1109/TSMC.1979.4310076

Referencias [1-3] citadas en la presentación principal.

Datos e implementación

[4] Leung, W. H., & Bovy, J. (2024). Galaxy10 DECaLS [Dataset]. Zenodo.

https://doi.org/10.5281/zenodo.10845026

[5] OpenCV. Image Thresholding. Referencia de implementación de Otsu.

docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html

Evidencia local
validacion_1000.csv · segmentacion.py · evaluar_qwen.py
config.json · predictions.csv · metrics.json · comparacion_mlflow.csv

Referencias [4-5] y evidencia local del experimento.

F1 por clase: siete tratamientos

Clase	Original	Gauss	CLAHE	G+Otsu	C+Otsu	G+S&M	C+S&M
0 · Disturbed	0.000	0.000	0.000	0.018	0.039	0.060	0.058
1 · Merging	0.510	0.510	0.628	0.447	0.613	0.361	0.037
2 · Round Smooth	0.321	0.336	0.388	0.332	0.388	0.352	0.284
3 · In-between Smooth	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4 · Cigar-shaped Smooth	0.296	0.249	0.290	0.275	0.287	0.174	0.195
5 · Barred Spiral	0.289	0.338	0.297	0.226	0.308	0.275	0.195
6 · Unbarred Tight Spiral	0.000	0.037	0.019	0.000	0.019	0.000	0.000
7 · Unbarred Loose Spiral	0.037	0.038	0.102	0.000	0.171	0.000	0.038
8 · Edge-on without Bulge	0.579	0.563	0.521	0.315	0.453	0.239	0.086
9 · Edge-on with Bulge	0.171	0.153	0.143	0.084	0.194	0.104	0.033

Cada clase tiene soporte de 100 imágenes. Los valores se redondean a tres decimales.

Prompt completo y configuración

Classify the morphology of the main galaxy
near the center of the image.

Choose exactly one category:

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

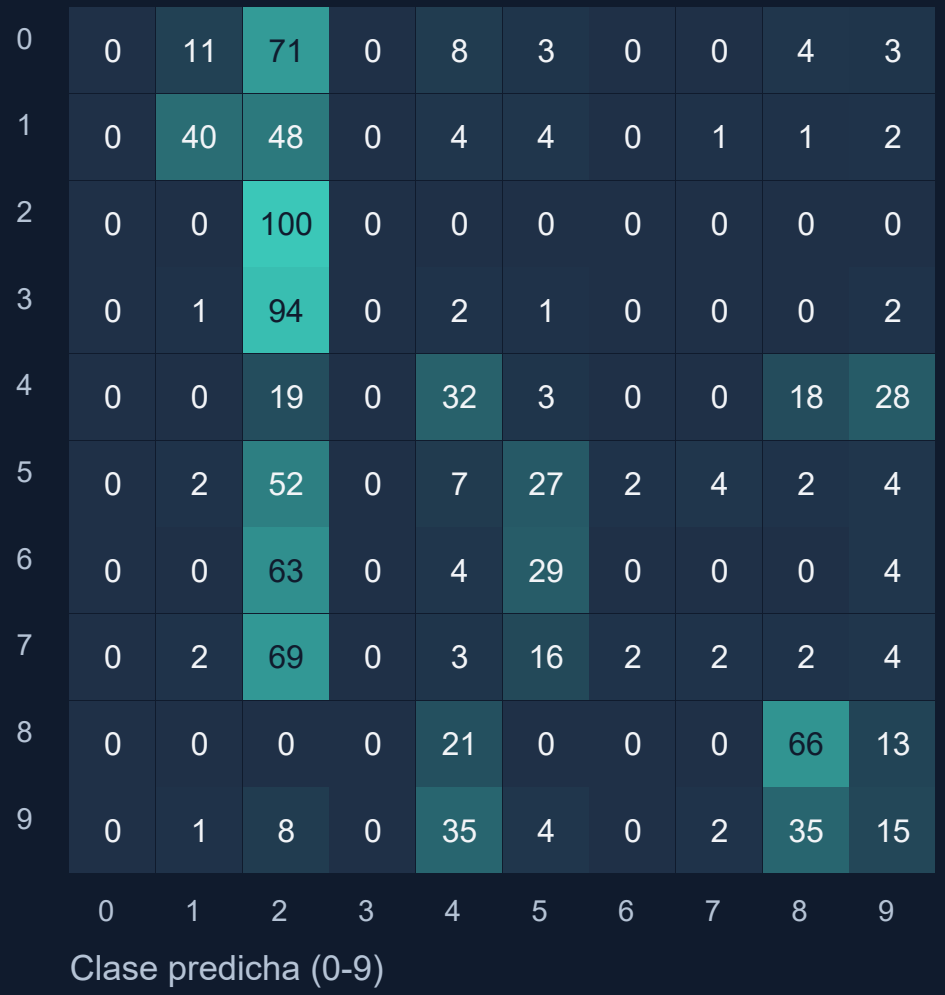
Return ONLY one digit from 0 to 9.
Do not explain.

Qwen3-VL-4B-Instruct
NF4 de 4 bits
do_sample = False
max_new_tokens = 16
seed = 42

Split and merge
tau = 10
min_size = 4
fondo_k = 3

Prompt exacto de evaluar_qwen.py. Misma configuración de inferencia en los siete tratamientos.

Matriz de confusión: Original

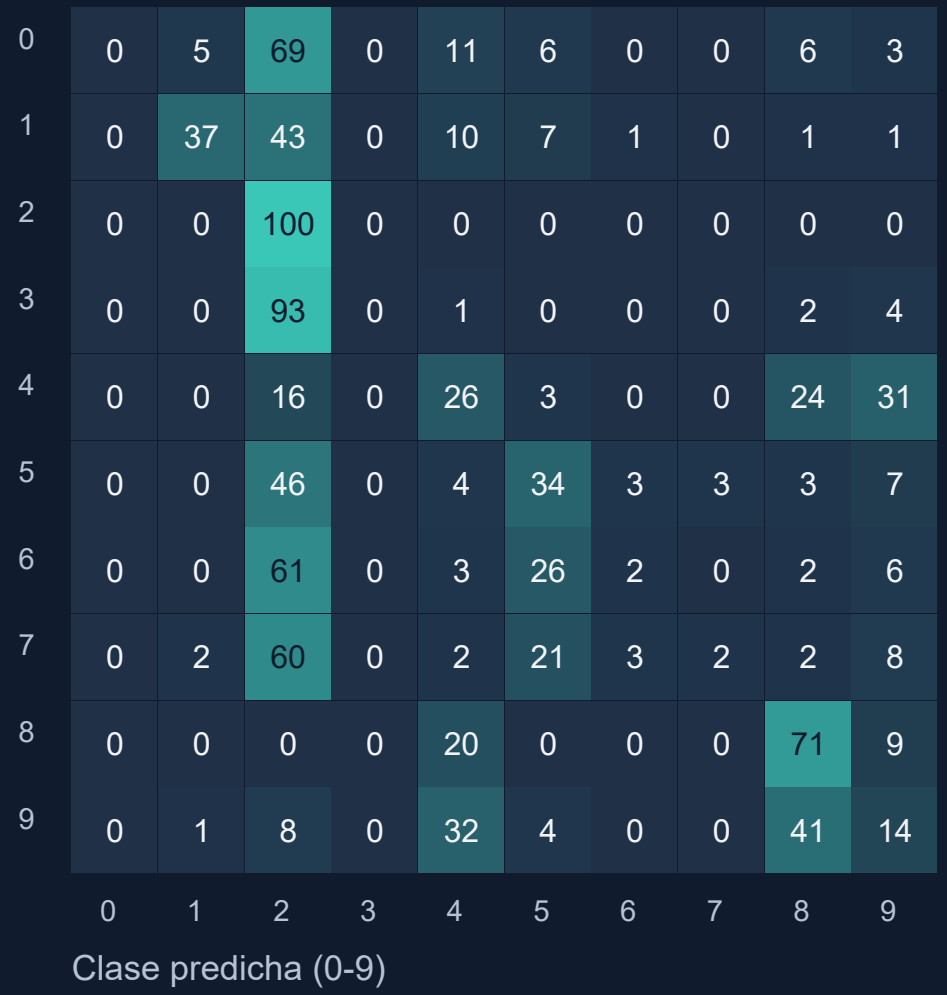


Accuracy: **28.2 %**
F1 macro: **0.2202**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: Gaussian



Accuracy: **28.6 %**

F1 macro: **0.2224**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: CLAHE

	0	1	2	3	4	5	6	7	8	9
0	0	13	61	0	11	4	0	1	8	2
1	0	60	25	0	9	3	1	1	1	0
2	0	0	100	0	0	0	0	0	0	0
3	1	0	84	0	4	1	0	0	1	9
4	0	0	5	0	32	2	0	1	28	32
5	0	9	36	0	6	30	3	4	6	6
6	0	1	46	0	1	37	1	5	4	5
7	0	6	53	0	2	21	2	6	3	7
8	0	0	0	0	24	0	0	0	68	8
9	1	2	6	0	32	4	0	0	42	13

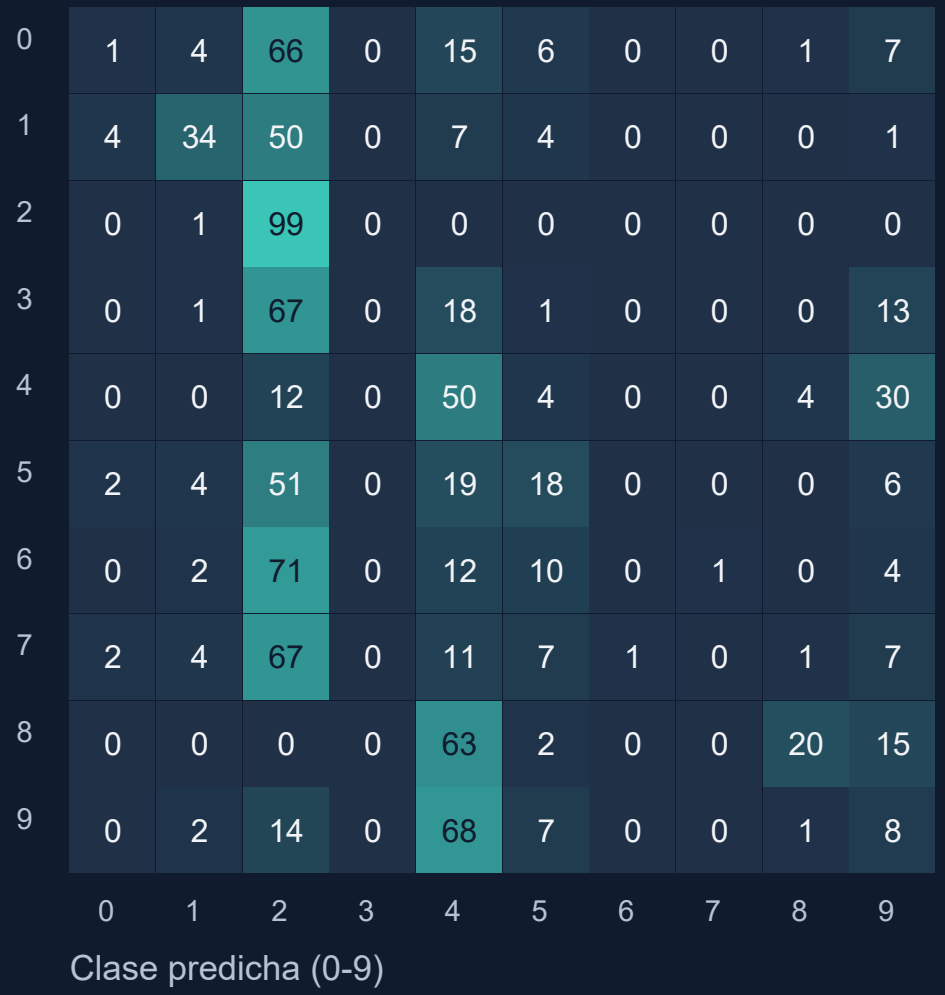
Clase predicha (0-9)

Accuracy: **31.0 %**
F1 macro: **0.2387**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: Gaussian + Otsu



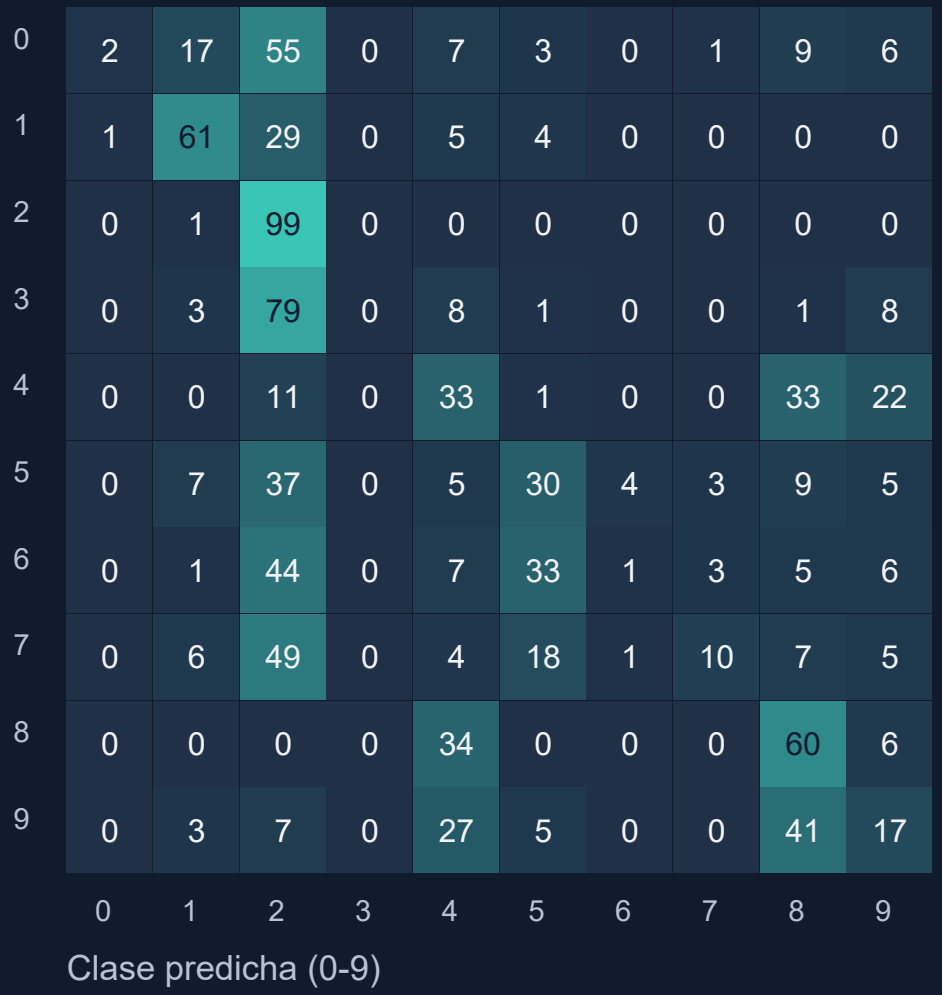
Accuracy: **23.0 %**

F1 macro: **0.1698**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: CLAHE + Otsu



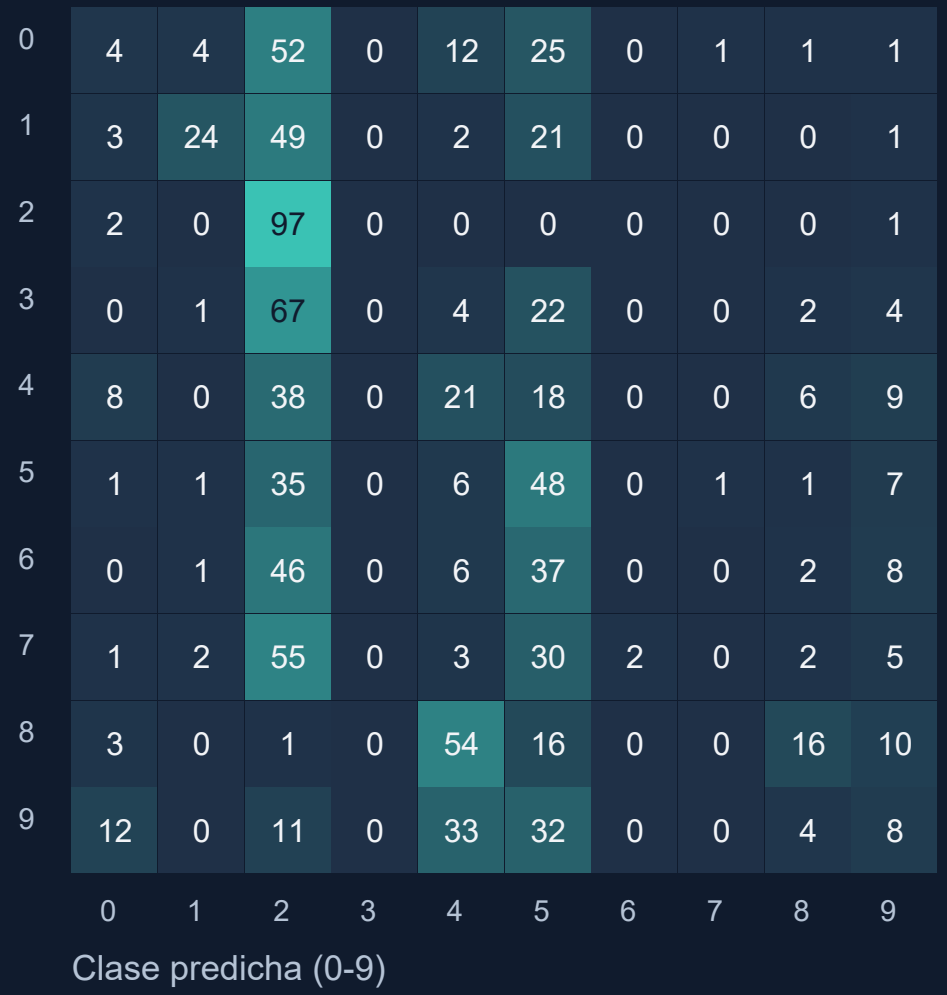
Accuracy: **31.3 %**

F1 macro: **0.2472**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: Gaussian + split and merge



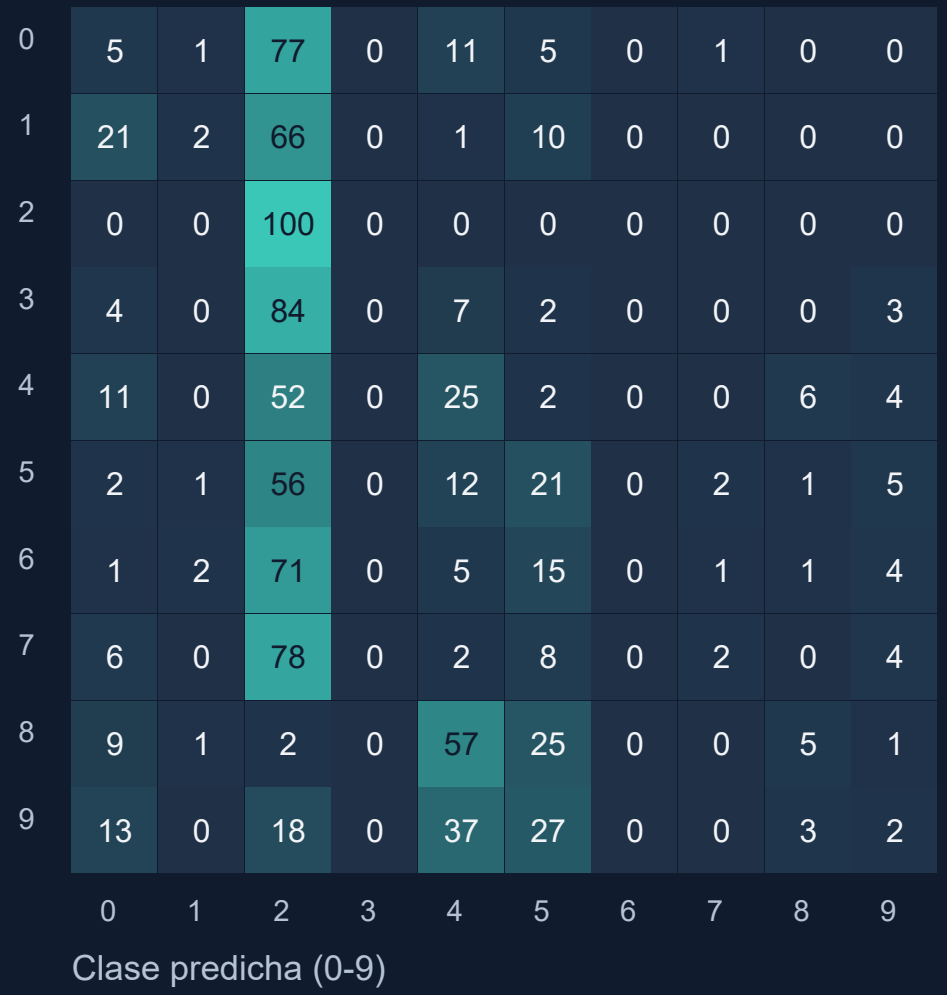
Accuracy: **21.8 %**

F1 macro: **0.1565**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.

Matriz de confusión: CLAHE + split and merge



Accuracy: **16.2 %**

F1 macro: **0.0926**

0: Disturbed
1: Merging
2: Round Smooth
3: In-between Smooth
4: Cigar-shaped Smooth
5: Barred Spiral
6: Unbarred Tight Spiral
7: Unbarred Loose Spiral
8: Edge-on without Bulge
9: Edge-on with Bulge

Filas: clase real. Columnas: clase predicha. Soporte de 100 imágenes en cada clase.